QP Code:

Reg. No: .....

Name: .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION, MARCH 2026

FOURTH SEMESTER

Discipline Specific Core Course (DSC)

MG4DSCFSQ201 – INTRODUCTION TO FOOD MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hrs **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions
Answer all questions
Each question carries **1** mark

1. Name the father of microbiology. [K]/[CO1]
   a) Louis Pasteur
   b) Antonie van Leeuwenhoek
   c) Robert Koch
   d) Paul Ehrlich

2. Discover the part of microscope that focuses light onto specimen. [U]/[CO1]
   a) Iris Diaphragm
   b) Condenser
   c) Objective lenses
   d) Eyepiece

3. Identify which of the following doesn't belongs to Eukaryote. [U]/[CO1]
   a) Bacteria
   b) Yeast
   c) Protozoa
   d) Mould

4. Recall microorganisms that use organic compounds as a source of both carbon and energy. [K]/[CO1]
   a) Autotroph
   b) Chemolithotroph
   c) Photoautotroph
   d) Chemoheterotroph

5. Select one of the following doesn't belongs to dry heat sterilization method. [K]/[CO2]

a) Flaming
b) Incineration
c) Hot Air Oven
d) Autoclave

6. Identify the cold sterilization method. [U]/[CO2]

a) Filtration
b) Radiation
c) Sonic Vibration
d) None of these

7. Indicate the holding period of autoclave. [U]/[CO2]

a) 100ºC for 20 minutes
b) 121ºC for 15 minutes
c) 160ºC for 2 Hours
d) 60ºC for 30 minutes

8. Recognize the characteristic of Botulinum toxin. [K]/[CO3]

a) Cytotoxin
b) Neurotoxin
c) Mycotoxin
d) Endotoxin

9. Select the most common cause of spoilage of milk at room temperature. [U]/[CO3]

a) Yeast
b) Mould
c) Lactic acid bacteria
d) Virus

10. Identify which food is most commonly associated with Salmonella food infection. [U]/[CO3]

a) Canned foods
b) Eggs and poultry
c) Acidic fruit juices
d) Sugary foods

[1 X 10 = 10]

Part B

Short Answer
Answer any **4** questions
Each question carries **5** marks

11. Differentiate between SEM and TEM. [U]/[CO1]
12. Describe the difference between Gram-positive and Gram-negative bacterial cell wall. [U]/[CO1]
13. Explain the morphology and structure of Fungi. [K]/[CO1]
14. Describe different methods of sterilization by heat. [K]/[CO2]
15. Explain the bacterial growth curve. [U]/[CO2]
16. Discuss on Bacillus cereus food infection. [U]/[CO3]

[5 X 4 = 20]

Part C

Long Essay

Answer any **2** questions

Each question carries **10** marks

17. Explain the structure and function of prokaryotic cell. [U]/[CO1]
18. Describe the culture methods used for the isolation of microorganisms. [U]/[CO2]
19. Discuss in detail on the biological spoilage of canned foods. [U]/[CO3]
20. Summarize the food infection and food intoxication with suitable examples. [U]/[CO3]

[10 X 2 = 20]